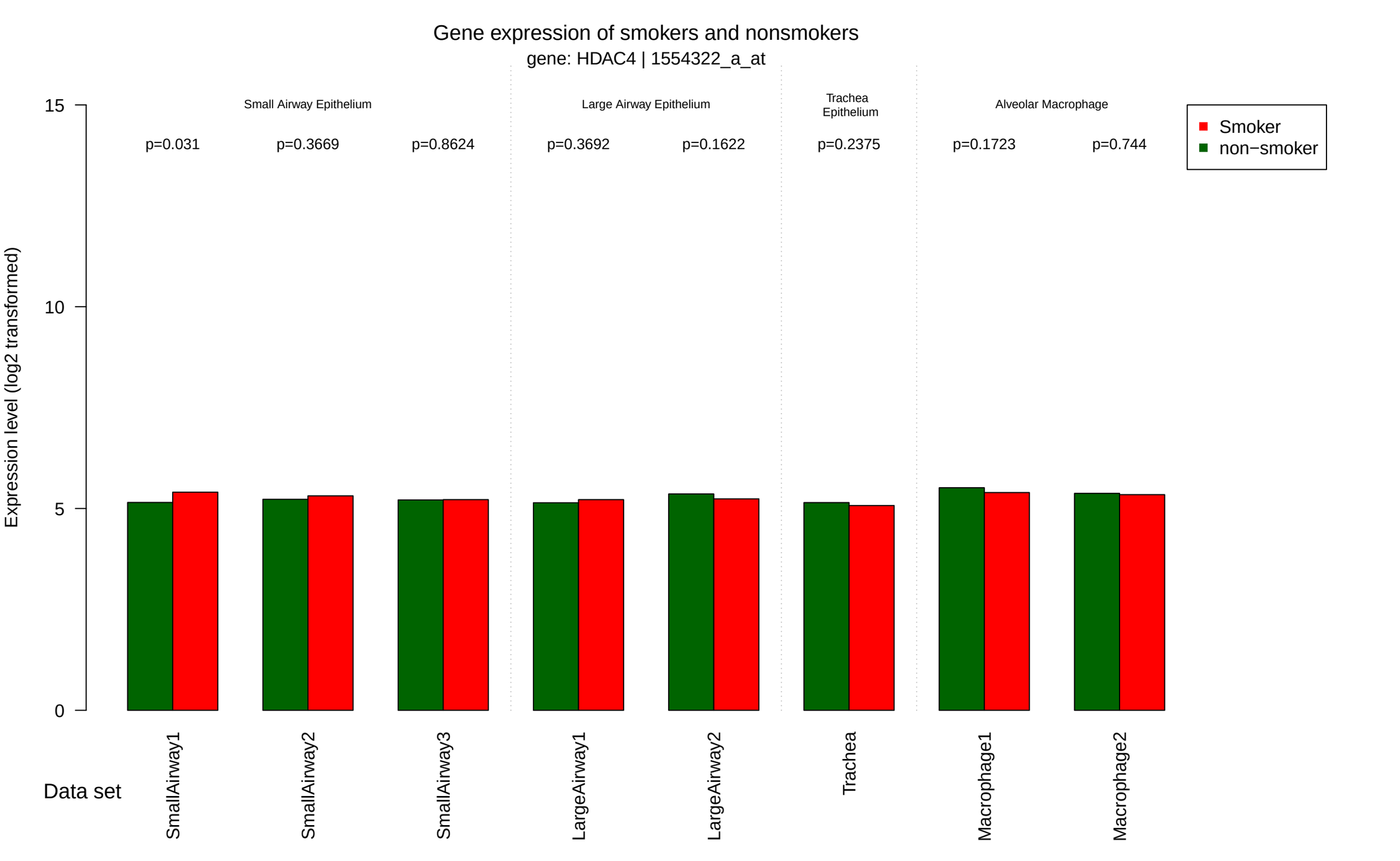
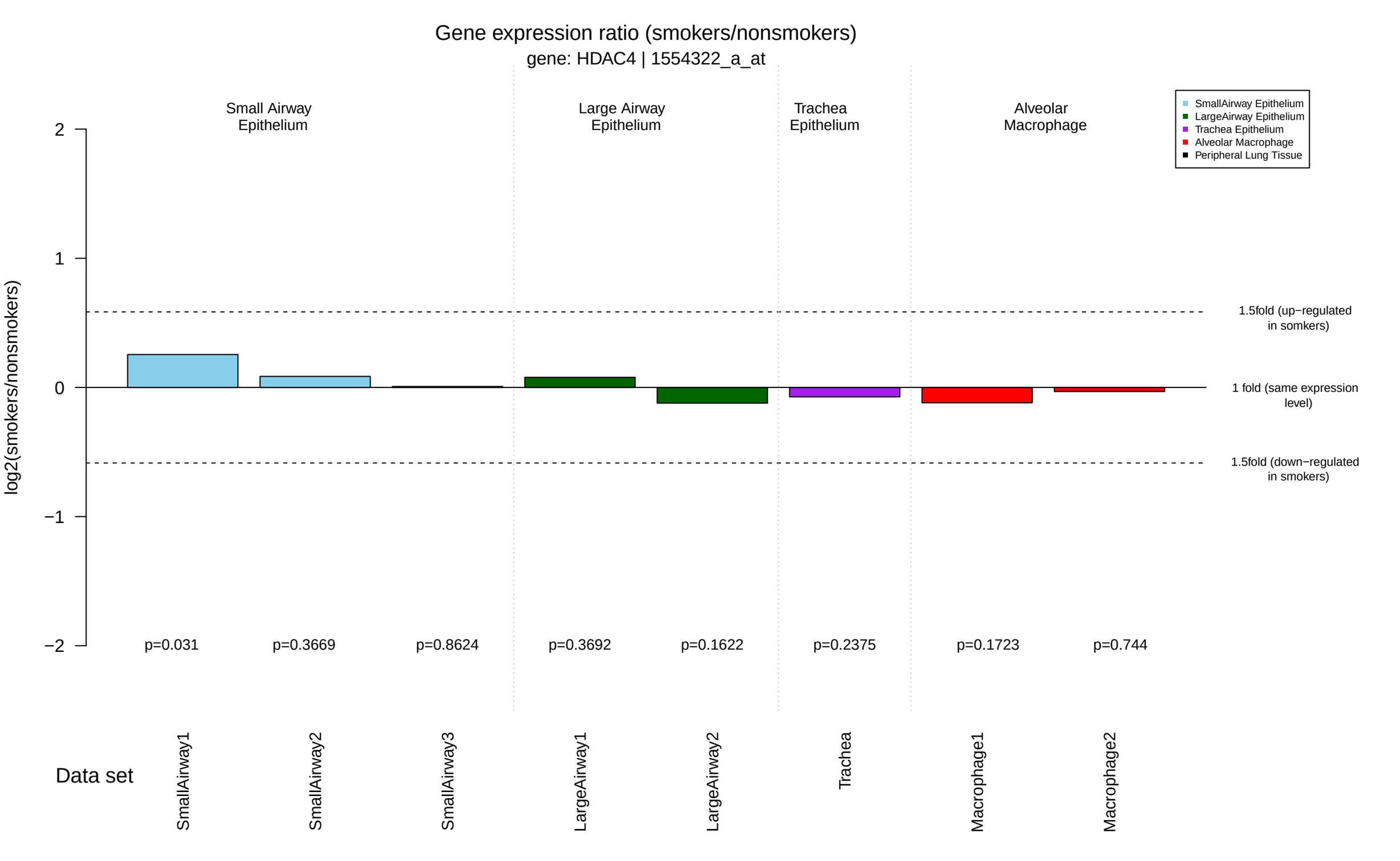
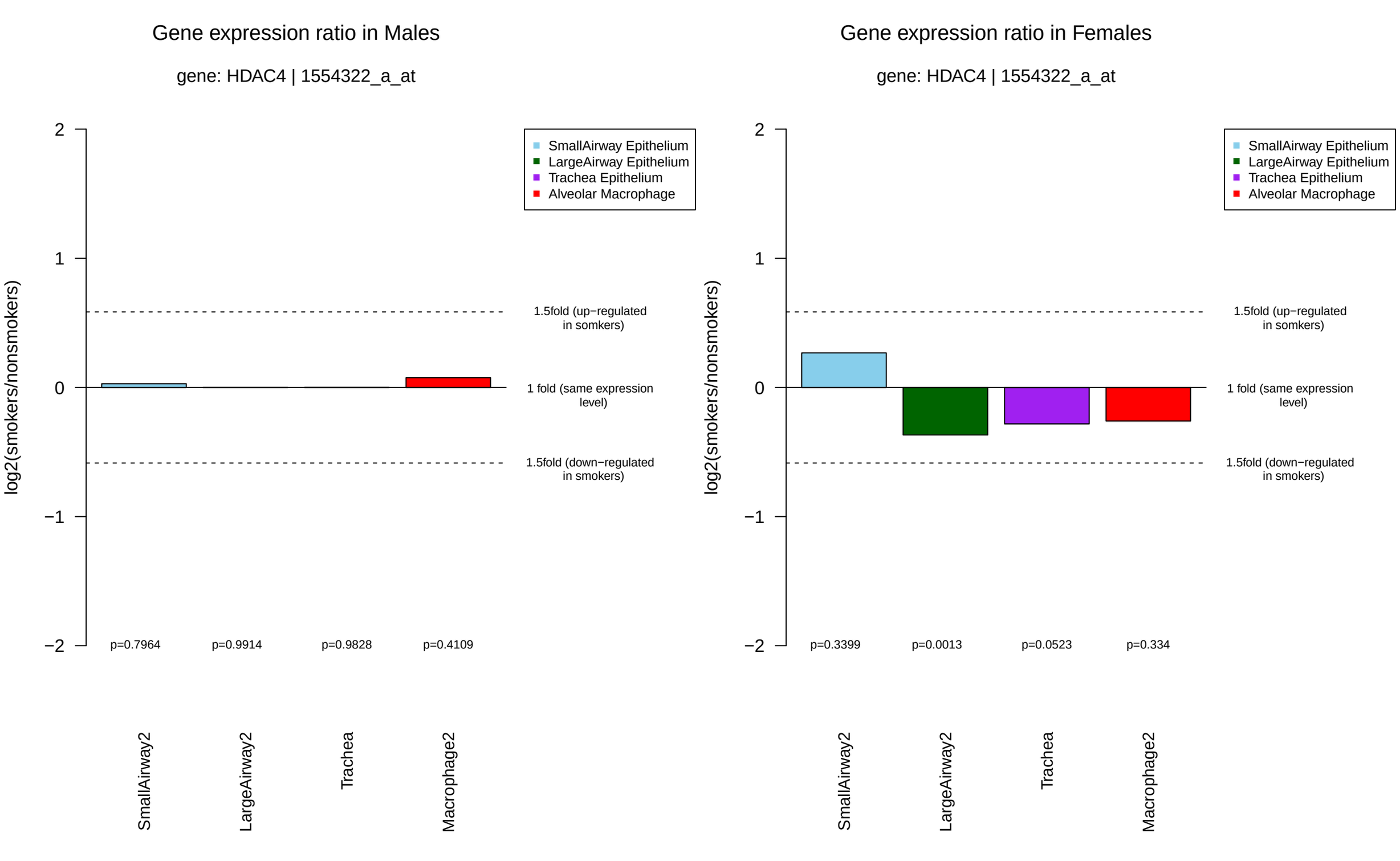
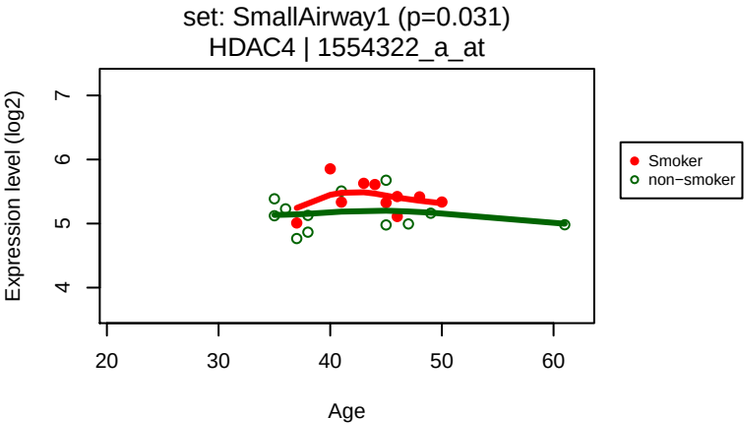
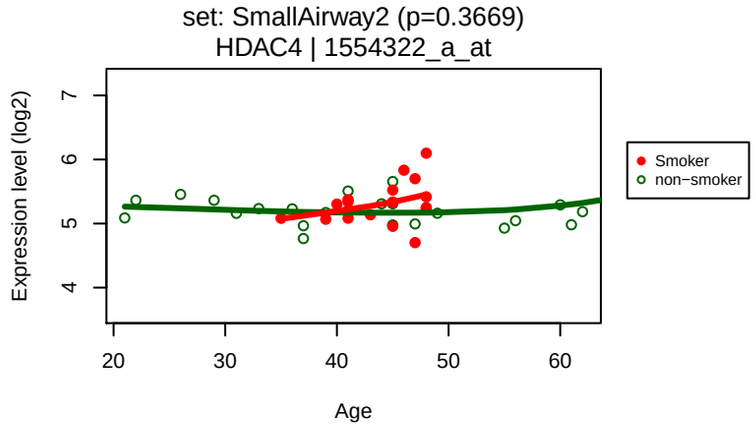
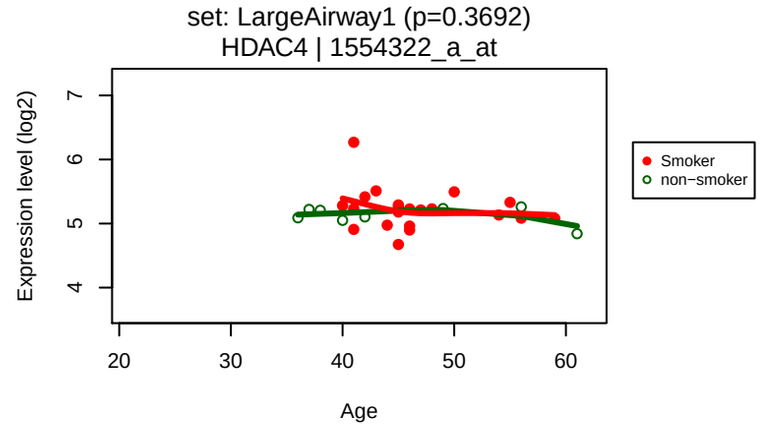

Gene expression of smokers and nonsmokers
gene: HDAC4 | 1554322_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.031
p=0.3669
p=0.8624
p=0.3692
p=0.1622
p=0.2375
p=0.1723
p=0.744
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

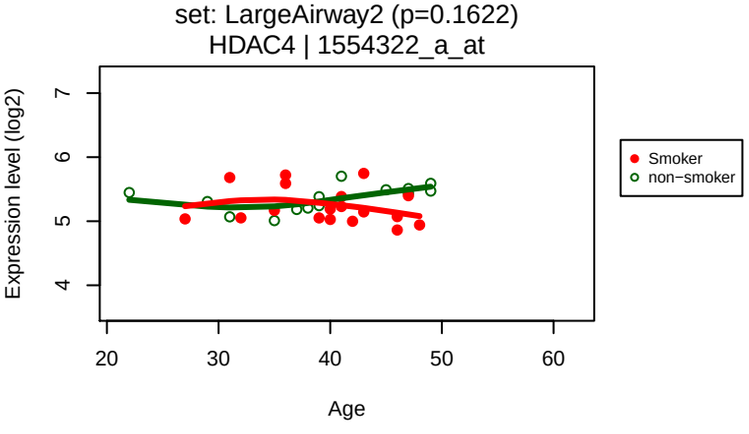

Gene expression ratio (smokers/nonsmokers)
gene: HDAC4 | 1554322_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.031
p=0.3669
p=0.8624
p=0.3692
p=0.1622
p=0.2375
p=0.1723
p=0.744
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

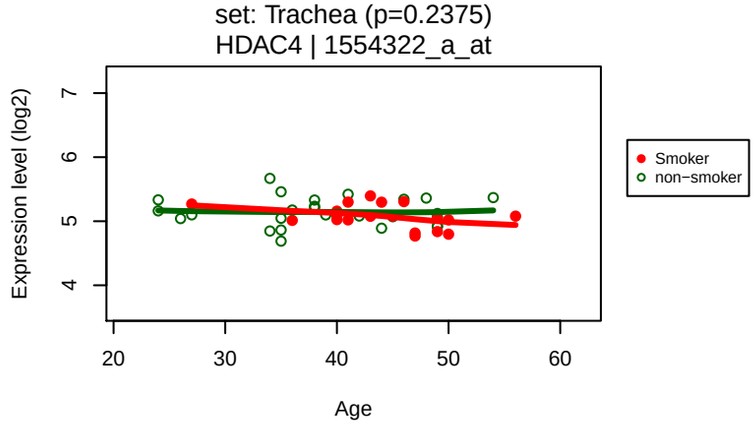

Gene expression ratio in Males
gene: HDAC4 | 1554322_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7964
p=0.9914
p=0.9828
p=0.4109
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HDAC4 | 1554322_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3399
p=0.0013
p=0.0523
p=0.334
SmallAirway2
LargeAirway2
Trachea
Macrophage2

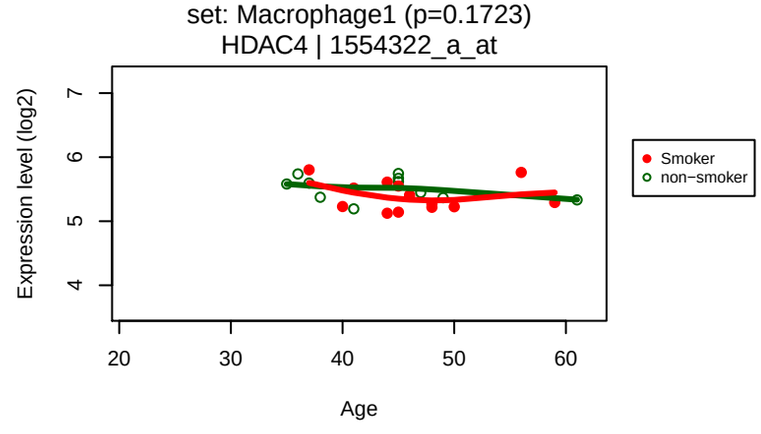

set: SmallAirway1 (p=0.031)
HDAC4 | 1554322_a_at
Expression level (log2)
Age
Smoker
non−smoker

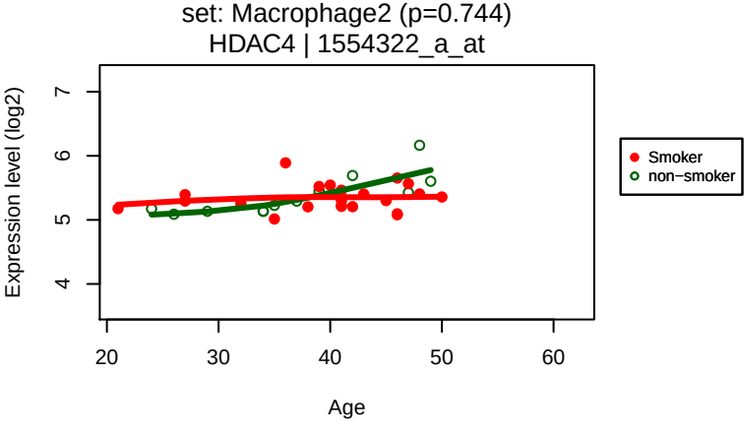

set: SmallAirway2 (p=0.3669)
HDAC4 | 1554322_a_at
Expression level (log2)
Age
Smoker
non−smoker

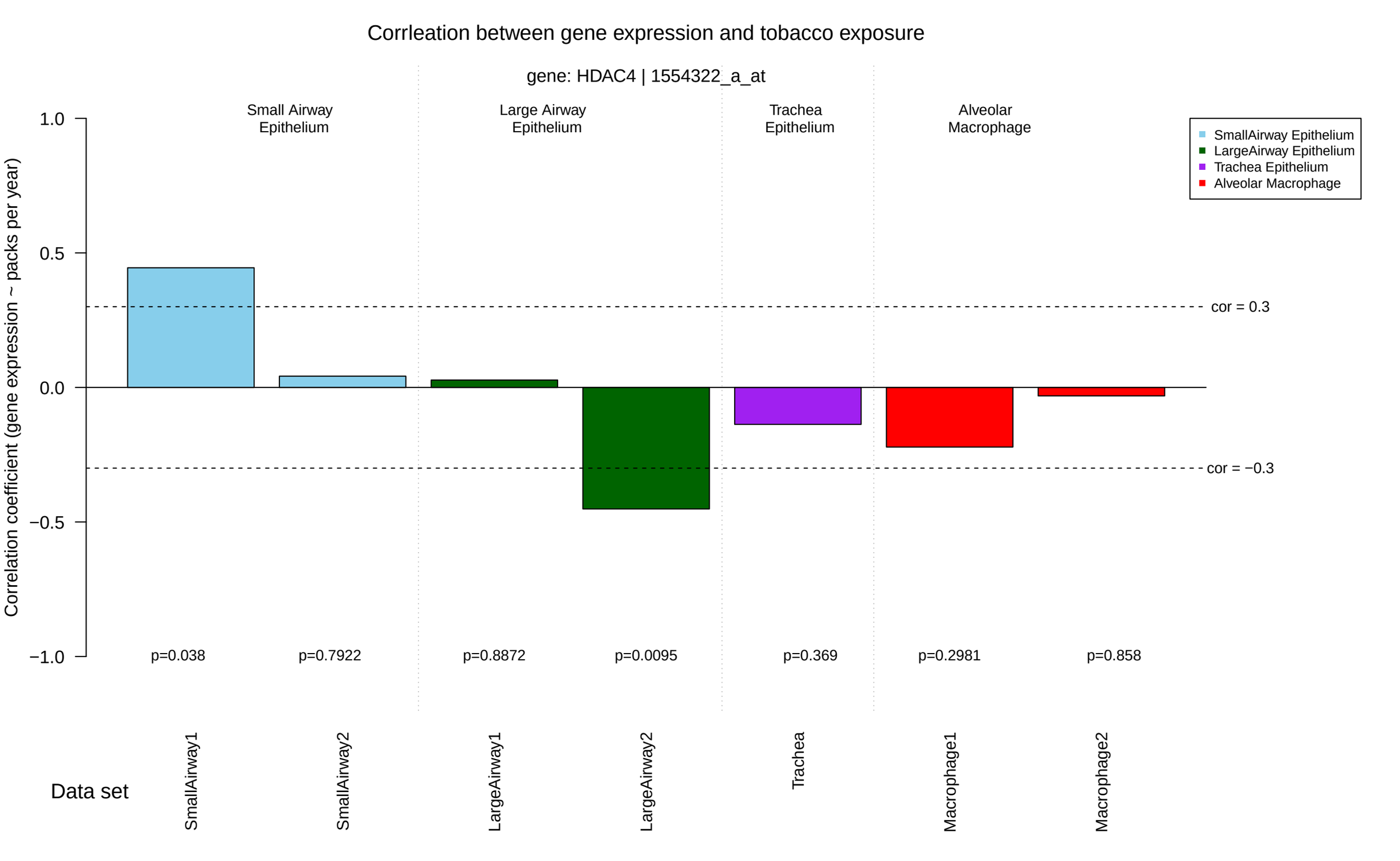

set: LargeAirway1 (p=0.3692)
HDAC4 | 1554322_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1622)
HDAC4 | 1554322_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2375)
HDAC4 | 1554322_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1723)
HDAC4 | 1554322_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.744)
HDAC4 | 1554322_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HDAC4 | 1554322_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.038
p=0.7922
p=0.8872
p=0.0095
p=0.369
p=0.2981
p=0.858
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2